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Non-fat Dry Milk & Skim Milk Powder

Product Definition

Nonfat dry milk and skimmed milk powder are very similar but are defined by two different sets of regulations and authorities. NDM is defined by the U.S. Food and Drug Administration, while SMP is defined by CODEX Alimentarius. Both are obtained by removing water from pasteurized skim milk. Both contain 5% or less moisture (by weight) and 1.5% or less milkfat (by weight). The difference is that skimmed milk powder has a minimum milk protein content of 34%, whereas nonfat dry milk has no standardized protein level.



Nonfat dry milk and skimmed milk powder are classified for use as ingredients according to the heat treatment used in their manufacture. There are three main classifications: high-heat, medium-heat and low-heat. Spray-dried nonfat dry milk and skimmed milk powders are available in two forms: ordinary or non-agglomerated (non-instant) and agglomerated (instant).

Typical Applications

For bakery, confectionery, dairy, meat products, and prepared mixes as:

- An economical source of nonfat dairy solids
- A source of functional dairy solids
- High-heat nonfat dry milk & skimmed milk powder is important for good loaf volume in breads
- Low-heat nonfat dry milk & skimmed milk powder is important for optimizing sensory properties in dairy foods and beverages
- A partial replacement for whey protein concentrate that provides similar concentrations of lactose but different types of protein and minerals/ash
- An easily transported and stored dairy ingredient

Typical Composition	
Protein	34.0% - 37.0%
Lactose	49.5% - 52.0%
Fat	0.6% - 1.25%
Ash	8.2% - 8.6%
Moisture	3.0% - 4.0% (non-instant) 3.5% - 4.5% (instant)

Physical and Chemical Characteristics

Typical Microbiological Analysis:	
Standard plate count	< 10,000 cfu/g*
Coliform	< 10/g (maximum)
E.coli	Negative
Salmonella	Negative
Listeria	Negative
Coagulase-positive staphylococci	Negative

Other Characteristics:	
Scorched particle content	7.5 - 15.0mg (spray-dried) 22.5mg (roller-dried)
Titratable acidity	0.14 - 0.15%
Solubility index	1.0 ml (instant) 1.2 ml (spray-dried, low-heat) 2.0 ml (spray-dried, high-heat) 15.0 ml (roller-dried)
Color	White to light cream color
Flavor	Clean, pleasing dairy flavor

*Extra grade

Packaging

Stitched or glued, multiwall kraft bag with polyethylene inner liner. No staples or metal fasteners. Also available in plastic-lined corrugated paperboard or aluminum tote bins.

Storage

Ship and store in a cool, dry environment at temperatures less than 27°C and relative humidity less than 65%. The shelf life of non-instant nonfat dry milk powder is 12-18 months; instant is 6-12 months. Note that storage life is very dependent on storage conditions, and that this figure is only a guide. Under ideal conditions, non-instant nonfat dry milk powder can retain its physical and functional properties for at least two years; however, quality will be impaired if temperatures and humidity are too high and storage is extended. For more information on milk powders, please see our [Milk Powder Storage Fact Sheet](#).

Recommended Uses of a Function of Heat Treatment

Typical Processing TreatmentUndenatured Whey Protein Nitrogen* (mg/g)Recommended Applications

Classification	Typical Processing Treatment	Undenatured Whey Protein Nitrogen* (mg/g)	Recommended Applications
Low-heat	Cumulative heat treatment of milk not more than 70°C for 2 minutes	> 6.00	Fluid milk fortification, cottage cheese, cultured skim milk, starter culture, chocolate dairy drinks, ice cream
Medium-heat	Cumulative heat treatment of 70 -78°C for 20 minutes	1.51 - 5.99	Prepared mixes, ice cream, confectionery, meat products
High-heat	Cumulative heat treatment of 88°C for 30 minutes	< 1.50	Bakery, meat products, ice cream, prepared mixes

* Higher temperatures and/or extended holding times contribute directly to whey protein denaturation. This index is used as a measure of the cumulative heat effects during processing of nonfat dry/skimmed milk powder.

